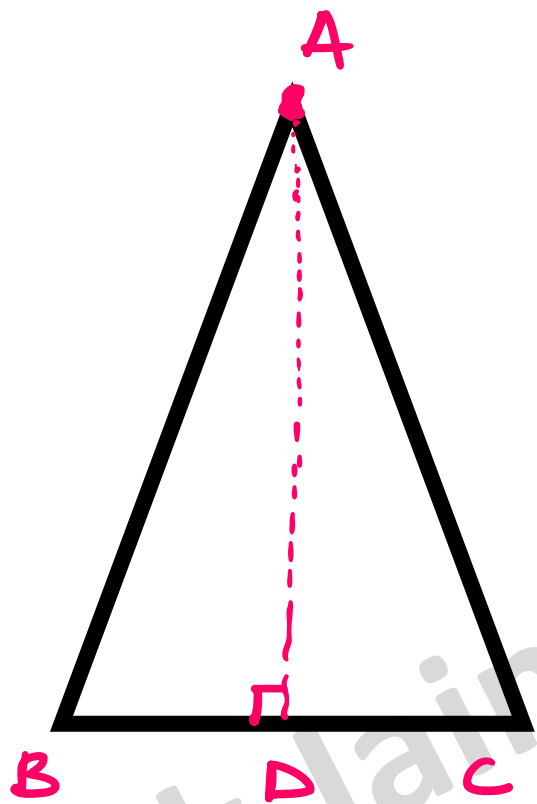
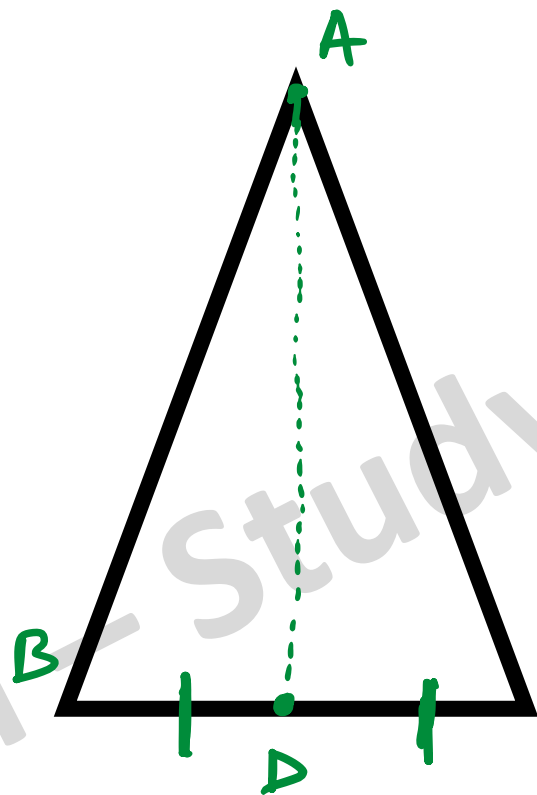




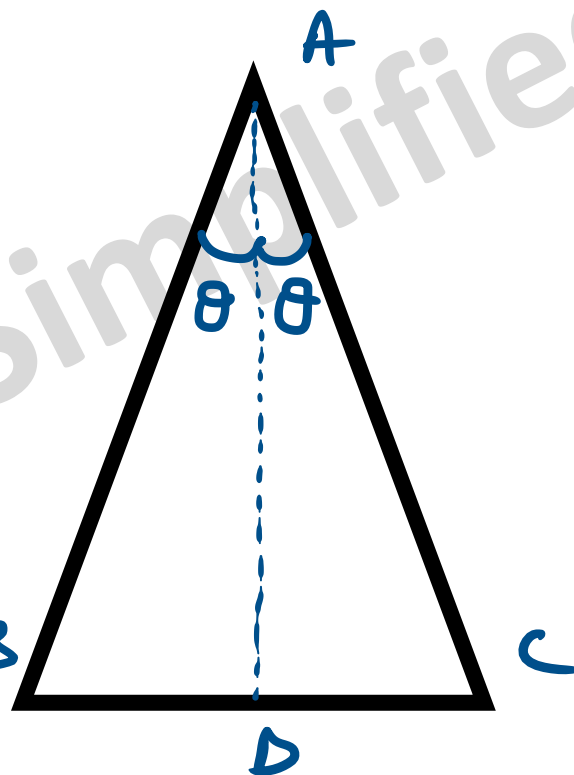
$AD = \text{altitude}$

$[\perp]$



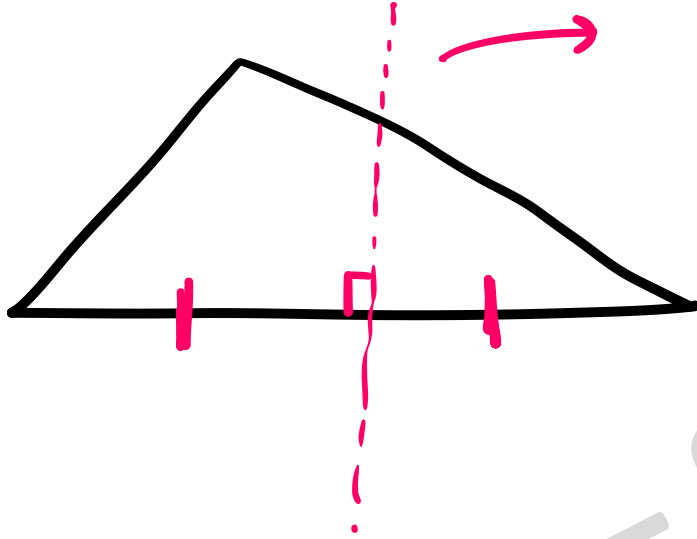
$AD = \text{Median}$

$[BD = DC]$



$AD = \text{angle bisector}$

$[\angle BAD = \angle DAC]$



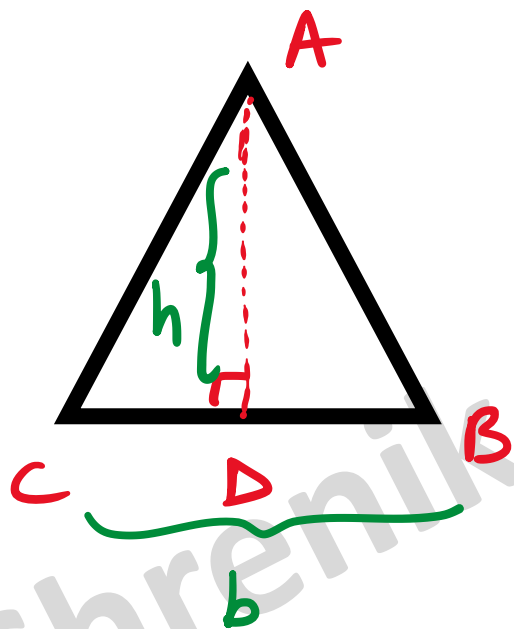
1'r Bisector

Shrenik Jain - Study Simplified

The sum of the base and altitude of a triangle is 30 cm.

What is the maximum possible area of such a triangle (in cm^2)

- (A) 100 (B) 110 (C) 112.5 (D) 120



①

$$b + h = 30$$

②

$$\begin{aligned} A(\Delta) &= \frac{1}{2}bh \\ &= \frac{1}{2}b(30-b) \end{aligned}$$

$$A = 15b - \frac{b^2}{2}$$

$$A = 15b - \frac{b^2}{2}$$

$$(3) \quad A = f(b) = \boxed{15b - \frac{b^2}{2}}$$

$$\rightarrow f'(b) = 0$$

$$\rightarrow \underline{15 - b} = 0$$

$$b = 15 \quad // \quad (\text{max pt})$$

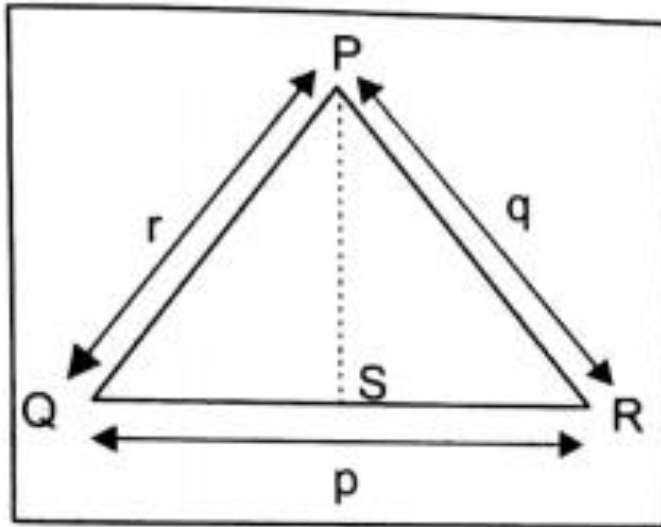
$$\rightarrow f''(b) = -1$$

$$f''(15) < 0 \quad \{\text{max}\}$$

$$A = f(b) = \boxed{15b - \frac{b^2}{2}}$$

$$A_{\max} = f(b=15) = 15(15) - \frac{(15)^2}{2} \\ = 112.5 \text{ cm}^2$$

In a triangle PQR, PS is the angle bisector of $\angle QPR$ and $\angle QPS = 60^\circ$. What is the length of PS?

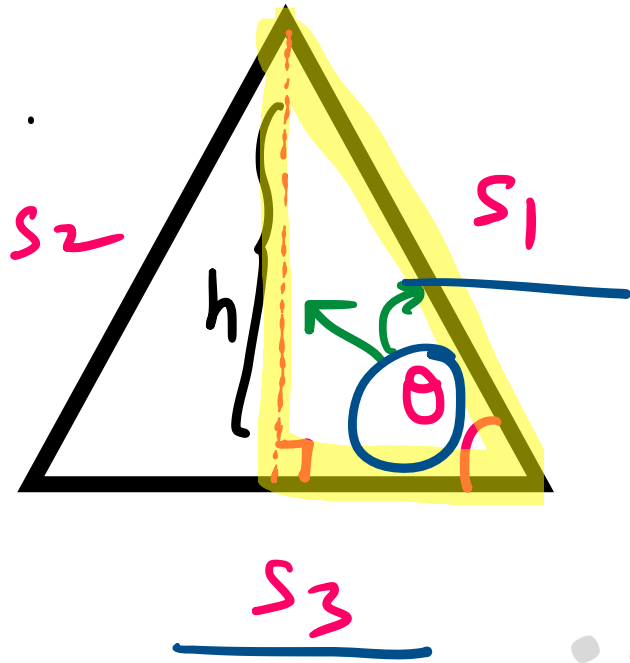


(a) $\frac{(q+r)}{qr}$

(b) $\frac{qr}{(q+r)}$

(c) $\sqrt{(q^2 + r^2)}$

(d) $\frac{(q+r)^2}{qr}$ [GATE-2015]



SAS

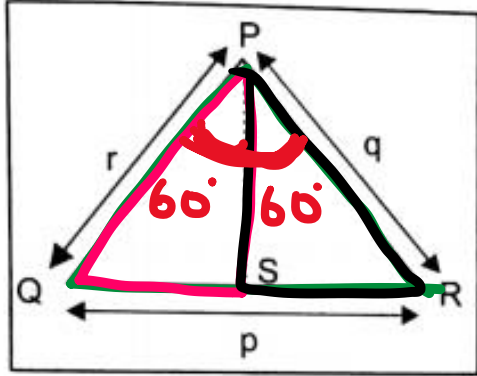
$$\begin{aligned} \textcircled{1} \quad A(\triangle) &= \frac{1}{2} b h \\ &= \frac{1}{2} (s_3) (\underline{h}) \end{aligned}$$

$$\textcircled{2} \quad \sin \theta = \frac{h}{s_1} \Rightarrow \underline{h = s_1 \sin \theta}$$

$$\textcircled{3} \quad A = \frac{1}{2} (s_3) \cdot (s_1 \sin \theta)$$

In a triangle PQR, PS is the angle bisector of $\angle QPR$ and $\angle QPS = 60^\circ$. What is the length of PS?

M1



(a) $\frac{(q+r)}{qr}$

(b) $\frac{qr}{(q+r)}$

(c) $\sqrt{(q^2 + r^2)}$

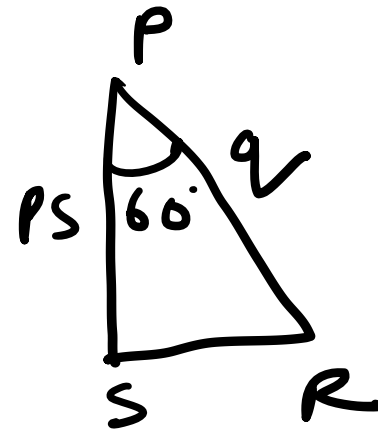
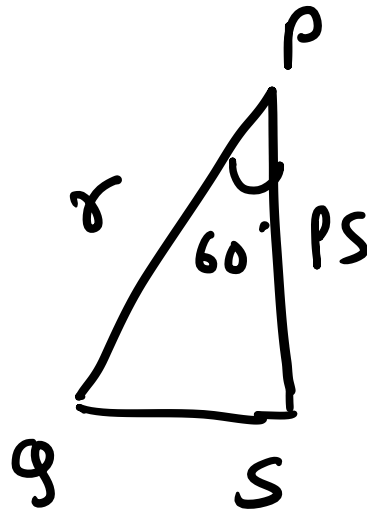
(d) $\frac{(q+r)^2}{qr}$ [GATE-2015]



SAS

① $A(\triangle PQR) = A(\triangle PQS) + A(\triangle PSR)$

② $\frac{1}{2} r \cdot q \cdot \sin 120^\circ = \frac{1}{2} r \cdot PS \sin 60^\circ + \frac{1}{2} PS \cdot q \sin 60^\circ$



$$r \cdot q \cdot \cancel{\sin 120}^{\sqrt{3}/2} = \cancel{r} \cdot \textcircled{PS} \cancel{\sin 60}^{\sqrt{3}/2} + \textcircled{PS} \cancel{q} \cancel{\sin 60}^{\sqrt{3}/2}$$

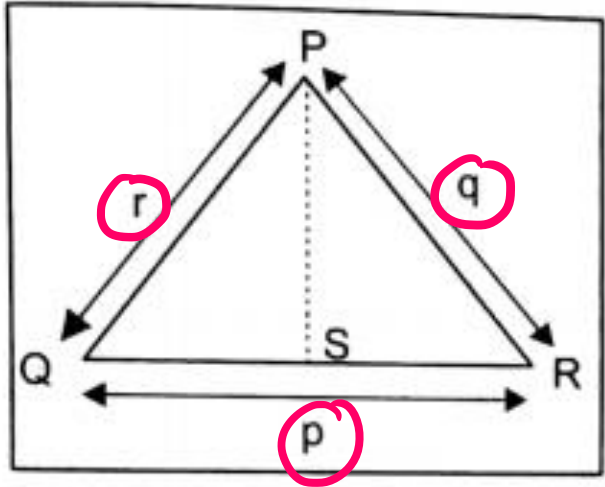
$$\Rightarrow r q = PS(r + q) \Rightarrow$$

$$PS = \frac{rq}{r+q}$$

$$\begin{aligned} \sin 120 &= \sin(90 + 30) \\ &= \cos 30 \\ &= \sqrt{3}/2 \end{aligned}$$

$$\sin 60 = \frac{\sqrt{3}}{2}$$

In a triangle PQR, PS is the angle bisector of $\angle QPR$ and $\angle QPS = 60^\circ$. What is the length of PS?



(M2)

① $r, p, q \Rightarrow \text{metre}$
[Substitution]

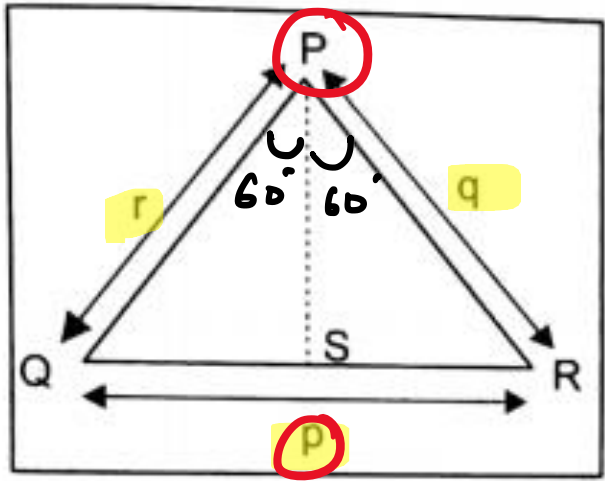
~~(a)~~ $\frac{(q+r)}{qr} = \frac{m}{m^2}$

$\checkmark$ (b) $\frac{qr}{(q+r)} = \frac{m^2}{m} = m$

$\checkmark$ (c) $\sqrt{(q^2 + r^2)}$
 $= \sqrt{m^2}$
 $= m$

~~(d)~~ $\frac{(q+r)^2}{qr}$ [GATE-2015]
 $\frac{m^2}{m^2}$

In a triangle PQR, PS is the angle bisector of $\angle QPR$ and $\angle QPS = 60^\circ$. What is the length of PS?

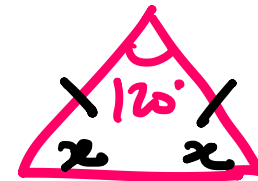
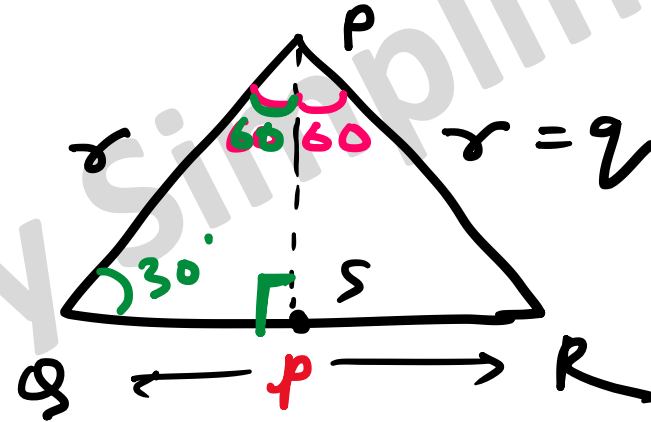


~~(a)~~ $\frac{(q+r)}{qr} = \frac{2r}{r^2}$
~~(b)~~ $\frac{qr}{(q+r)} = \frac{r^2}{2r}$
~~(c)~~ $\sqrt{(q^2 + r^2)}$
~~(d)~~ $\frac{(q+r)^2}{qr}$ [GATE-2015]

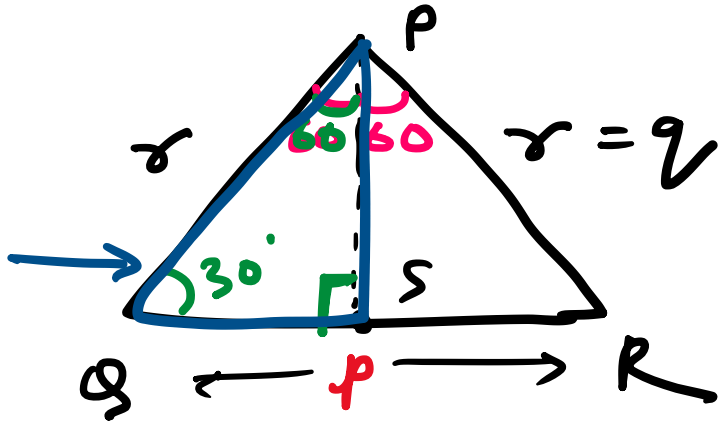
$= \sqrt{r^2 + r^2}$
 $= \sqrt{2r^2} = \sqrt{2}r$

①

$r = q$



$\Rightarrow 120 + 2x = 180$
 $x = 30^\circ$



$$\sin 30^\circ = \frac{PS}{r}$$

$$PS = \frac{1}{2} \cdot r = \frac{r}{2} //$$

∴ The smallest angle of a triangle is equal to the $\frac{2}{3}$ of the smallest angle of quadrilateral. (The ratio between the angle of quadrilateral is $3:4:5:6$.) The largest angle of the triangle is twice its smallest angle. What is the sum of the second largest angle of the triangle and largest angle of quadrilateral? [CE 2014]

① Angles of quadrilateral = $3k, 4k, 5k, 6k$

② $3k + 4k + 5k + 6k = 360 \rightarrow k = 20^\circ$

③ Largest angle of quadrilateral = $6k$
 $= 120^\circ //$

∴ → Smallest angle of Δ

$$= \frac{2}{3} \times 3k$$

$$= 2k$$

$$= 2 \times 20 = 40^\circ //$$

→ Largest angle of $\Delta = 40 \times 2 = 80^\circ //$

→ $40 + 2^{\text{nd}} \angle + 80^\circ = 180^\circ$

2nd largest $\angle = 60^\circ$

$$\text{Ans} = 60^\circ + 120^\circ = 180^\circ //$$

Shrenik Jain — Study Simplified

. Quadrilateral



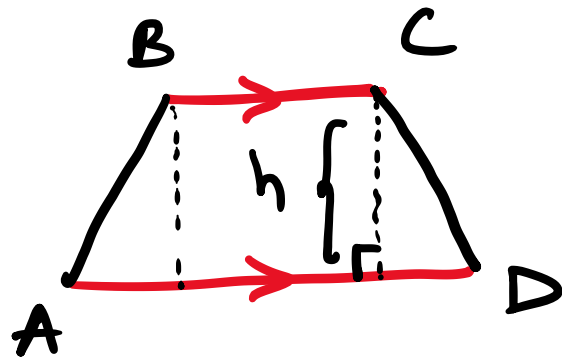
4 sided polygon



Σ of interior angle = 360°

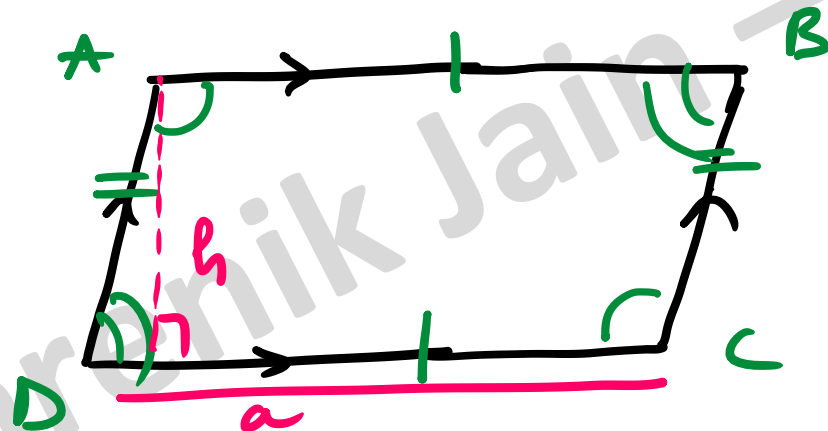


Σ of exterior angle = 360°



Trapezoid

$$\rightarrow A = \frac{(BC + AD)}{2} \times h$$



Parallelogram

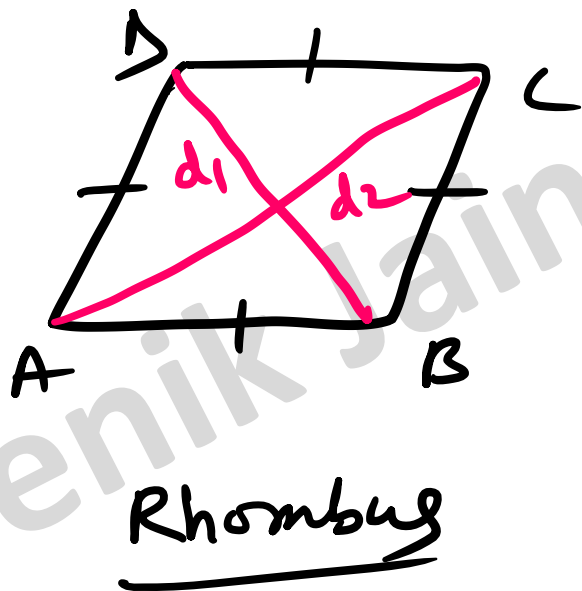
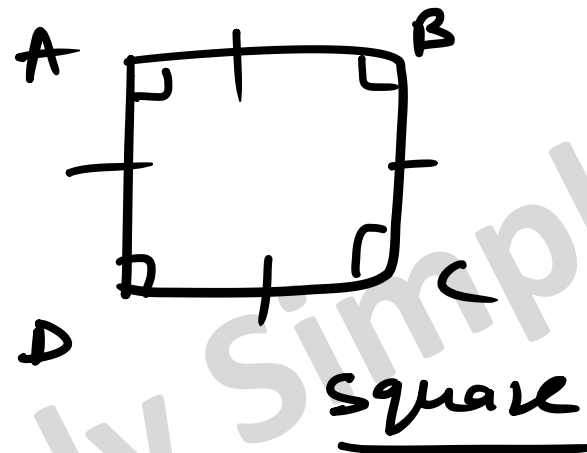
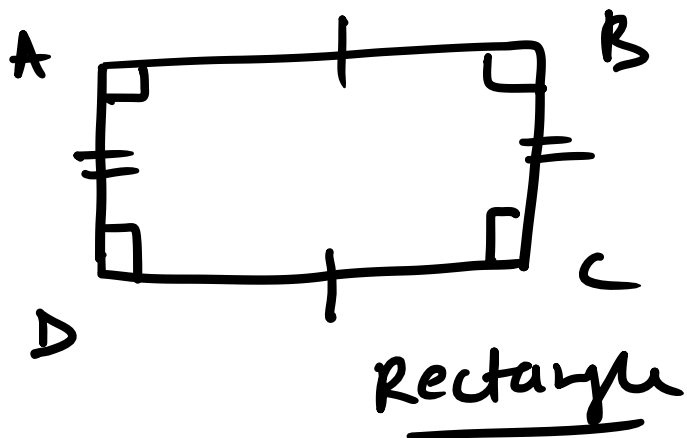
$$1) AB \parallel CD$$

$$AD \parallel BC$$

$$2) AB = CD, AD = BC$$

$$3) \angle A = \angle C$$

$$\angle B = \angle D$$



Shapes

Perimeter

Area

① Rectangle

$$2(l+b)$$

$$lb$$

② Square

$$4s$$

$$s^2$$

③ Parallelogram

$$2(a+b)$$

$$ah$$

④ Rhombus

$$4a$$

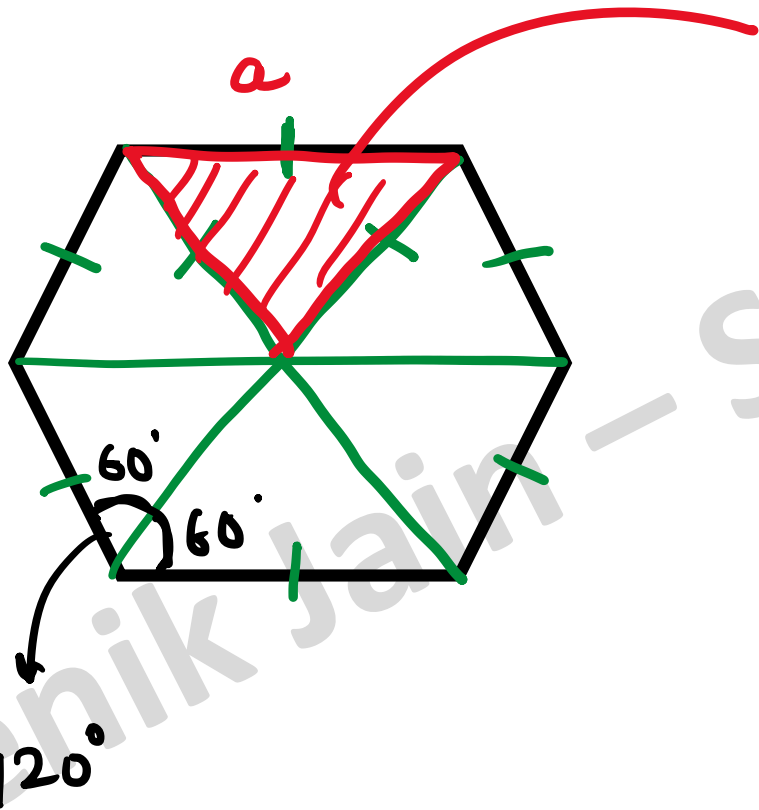
$$\frac{1}{2} d_1 d_2$$

⑤ Trapezium

Sum of
all sides

$$\frac{1}{2} h (a+b)$$

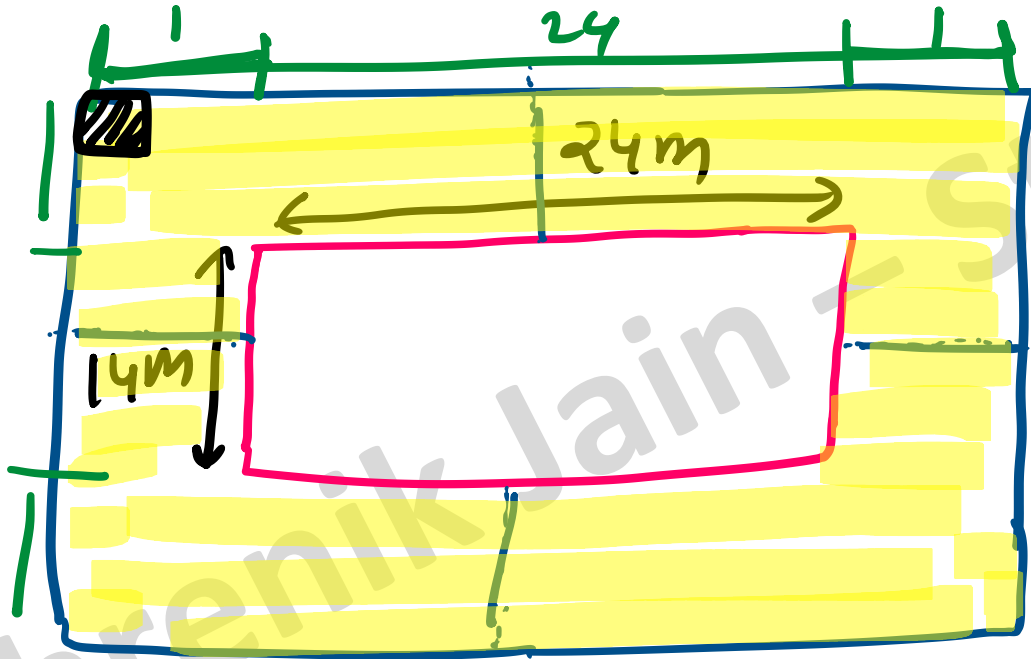
* Regular Hexagon



$$A (\text{Equilateral } \triangle) = \frac{\sqrt{3}}{4} a^2$$

$$\begin{aligned} \therefore A (\text{Regular Hexagon}) &= 6 \times A(\triangle) \\ &= 6 \times \frac{\sqrt{3}}{4} a^2 \\ &= \frac{3\sqrt{3}}{2} a^2 // \end{aligned}$$

- A garden is 24m long and 14m wide. There is a path 1m wide outside the garden along its sides. If the path is to be constructed with square marble tiles 10cmx10cm, then find the number of tiles required to cover the path.



$$\textcircled{1} A(\square) = 24 \times 14 = 336 \text{ m}^2 //$$

$$A(\square + \text{path}) = (26)(16) \\ = 416 \text{ m}^2 //$$

$$\Rightarrow A(\text{path}) = 416 - 336 \\ = 80 \text{ m}^2$$

$$\cdot A(\text{tile}) = 10\text{cm} \times 10\text{cm} = 100\text{ cm}^2$$

$$= 100 \left(\frac{1}{100} \text{ m} \right)^2$$

$$= \frac{100}{10000} \text{ m}^2$$

$$= \frac{1}{100} \text{ m}^2$$

$$1\text{m} = 100\text{cm}$$

$$1\text{cm} = \frac{1}{100} \text{ m}$$

$$\text{No. of tiles} = \frac{80 \cancel{\text{m}^2} \text{ (total Area)}}{\frac{1}{100} \cancel{\text{m}^2} \text{ (Area/1 tile)}}$$

$$= 80 \times 100 \text{ total tiles}$$

$$= 8000 //$$

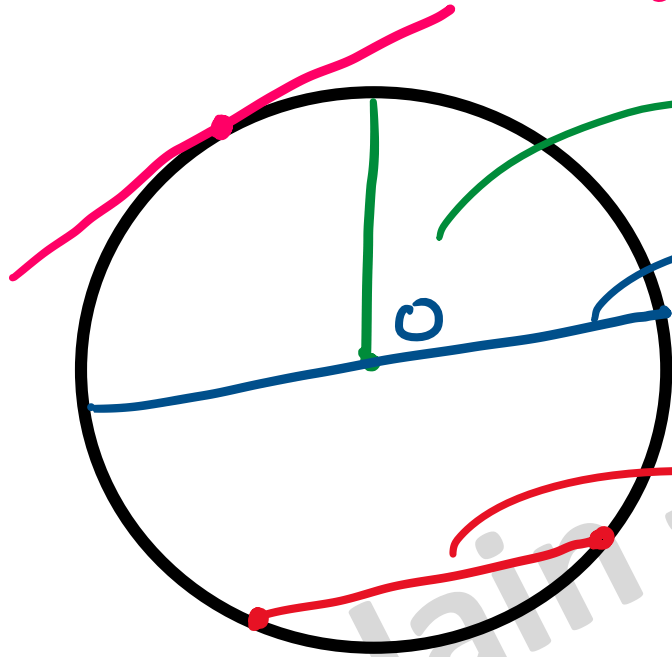
CIRCLE

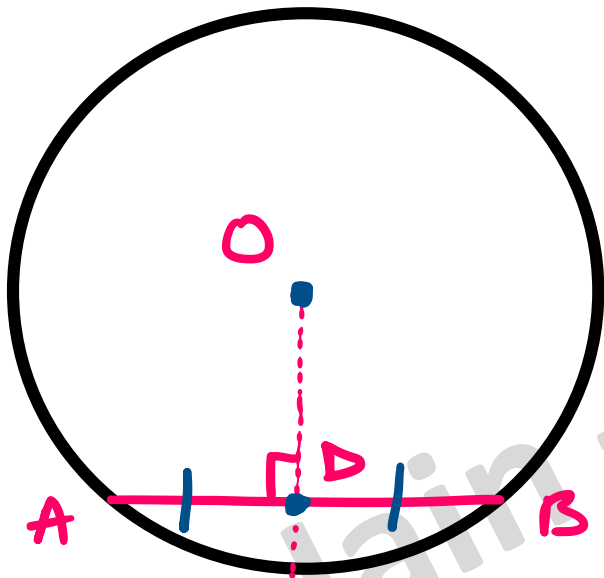
tangent

Radius (r)

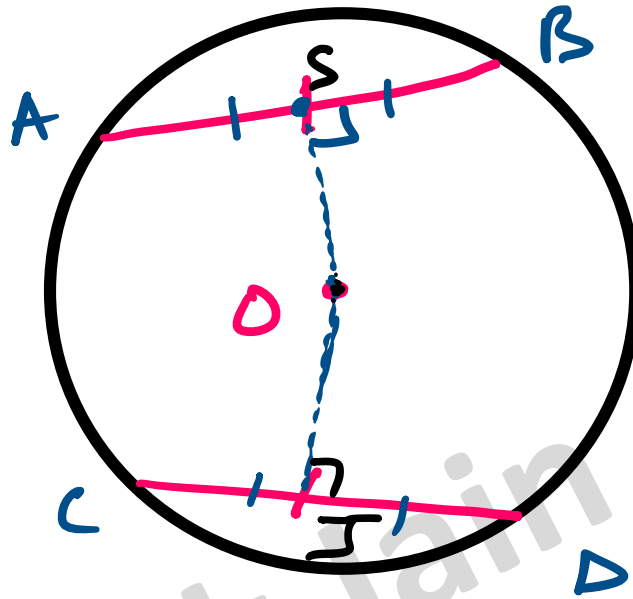
Diameter (d)

chord





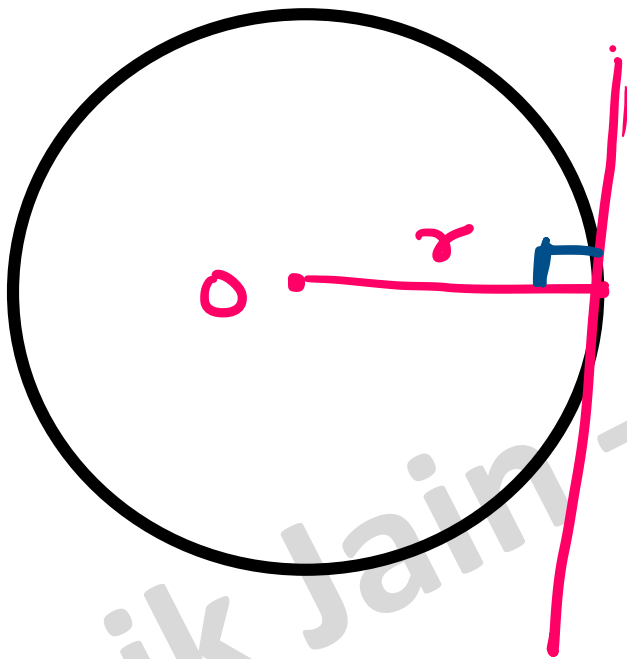
- 1) $\perp$ r from centre to chord bisects the chord
- 2) Line from centre to midpoint of chord is $\perp$ r to chord
- 3) $\perp$ r bisector of a chord passes through centre of circle.

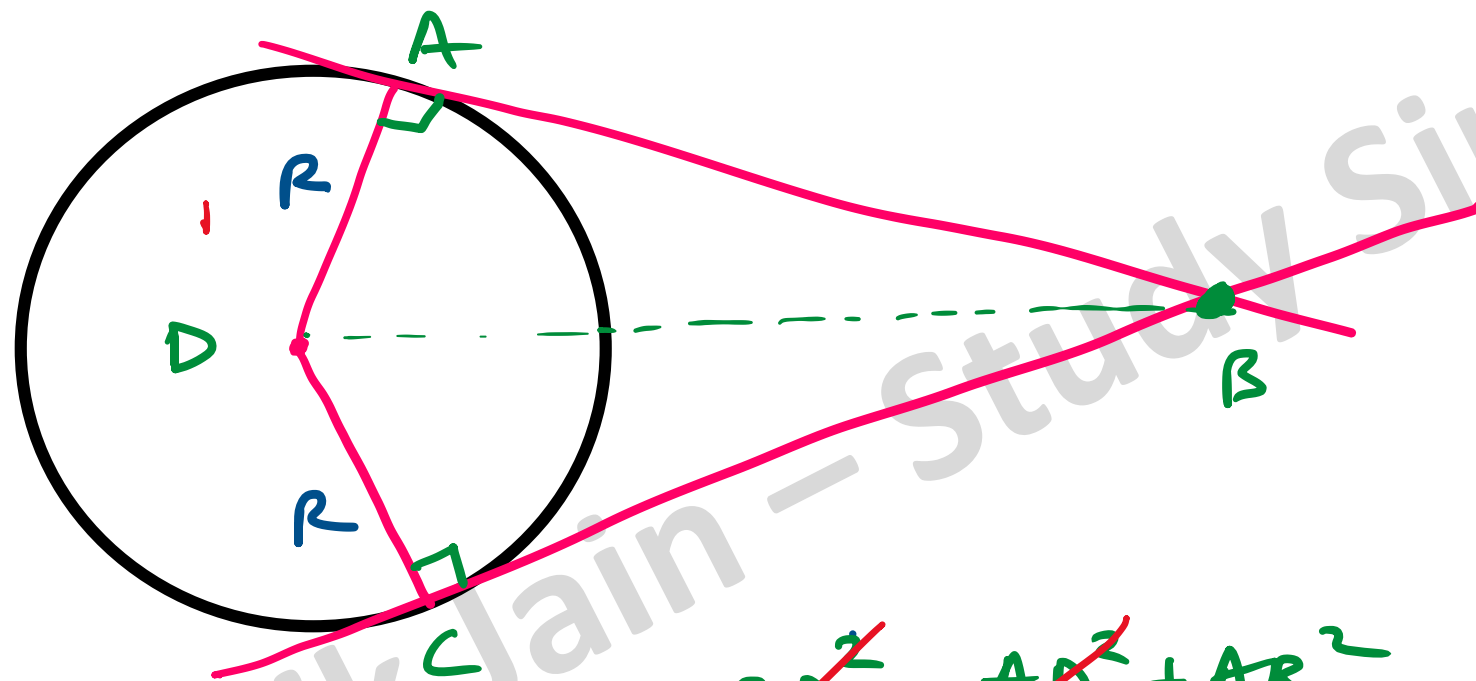


$$AB = CD$$

$$OS = OJ$$

- (i) Equal chords in equal circle are equidistant from centre



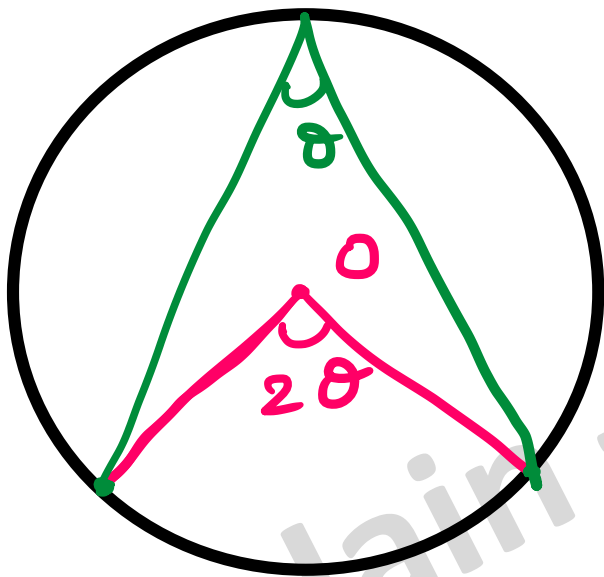


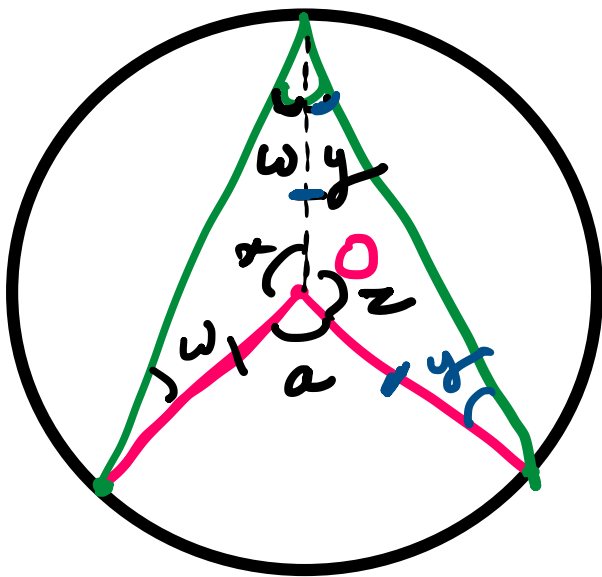
$$AB = BC$$

~~$$BD^2 = AD^2 + AB^2$$~~

~~$$BD^2 = DC^2 + BC^2$$~~

$$AB^2 - BC^2 = 0$$





$$1) \quad x + z + a = 360^\circ$$

$$2) \quad w + x + w = 180^\circ$$

$$x = 180^\circ - 2w$$

$$3) \quad z = 180^\circ - 2y$$

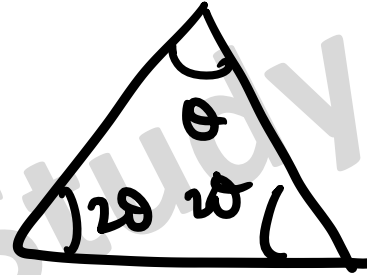
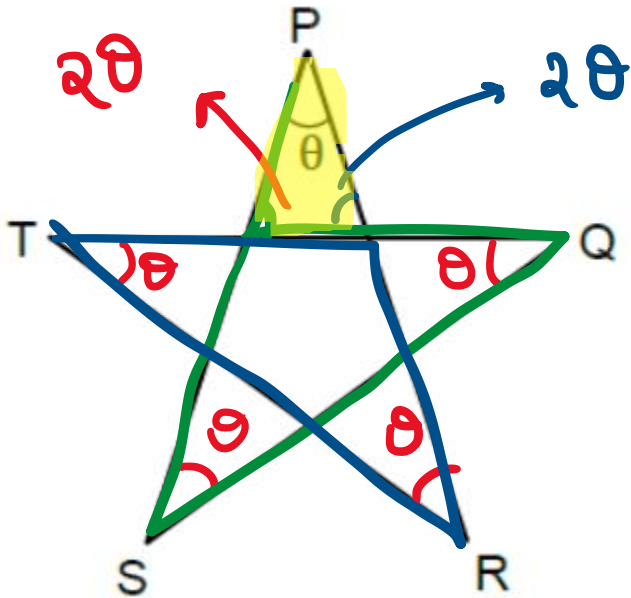
$$\Rightarrow a = 360 - x - z$$

$$= 360 - (180 - 2w) - (180 - 2y)$$

$$a = 2w + 2y = 2(w + y)$$

Five line segments of equal lengths PR, PS, QS, QT, RT are used to form a star as shown in the figure above.

Value of θ in degree is **[2021]**



$$\theta + 2\theta + 2\theta = 180^\circ$$

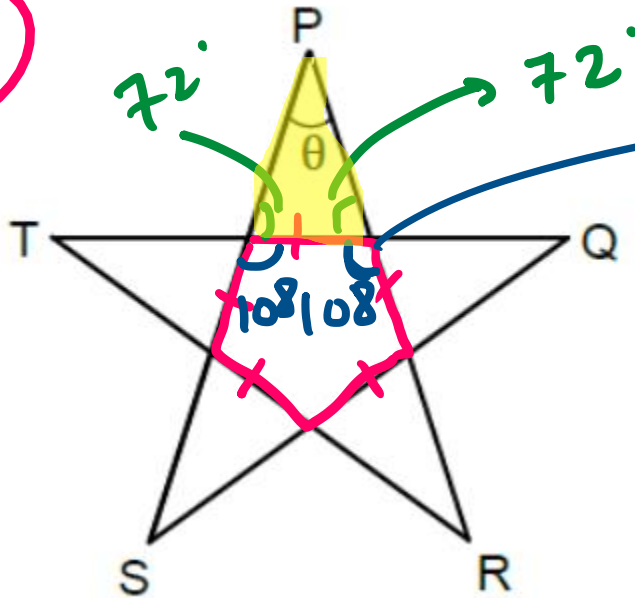
$$\theta = \frac{180}{5} = 36^\circ //$$

M1

Five line segments of equal lengths PR, PS, QS, QT, RT are used to form a star as shown in the figure above.

Value of θ in degree is [2021]

M2



$$72 + 72 + \theta = 180^\circ$$

$$\theta = 36^\circ //$$

① Regular pentagon

→ Σ of Interior angles

$$= (n-2)180$$

$$= (5-2)180$$

$$\rightarrow \text{Each interior angle} = \frac{(5-2)180}{5}$$

$$= 108$$